SMART SOLAR WIRELESS EV CHARGING STATION WITH IOT MONITORING

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CERTIFICATE

This is to certify that this is the project report entitled **"SMART SOLAR WIRELESS EV CHARGING STATION WITH IOT MONITORING"** submitted by **K.ABHINAYA (2411EC010075), K.SUBHASH (2411EC010077), K.VIKAS (2411EC010079), S.ASHA SAMEERA (2411EC010095), N.RISHIKA (2411EC010110), G.HARISH (2411EC010113) and R.SHIVA KUMAR (2411EC010117)** B. Tech II year II semester, Department of Electronics and Communication Engineering(ECE), during the year 2025-26. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project report **"SMART SOLAR WIRELESS EV CHARGING STATION WITH IOT MONITORING",** has been carried out by us and this work has been submitted to the **Department of Electronics and Communication Engineering (ECE), Malla Reddy University, Hyderabad**. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

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ABSTRACT

Nowadays, the use of electric vehicles is increasing rapidly, which creates a need for better and smarter charging systems. Most of the existing charging methods use wired connections and depend on electricity from the grid. This leads to problems like energy loss, lack of flexibility, and dependency on non-renewable energy sources. In this project, we developed a smart EV charging system that uses solar energy and wireless power transfer. The solar panel is used to generate electricity, which is stored in a battery. This stored energy is then used for charging the vehicle without using any physical wires. The main controller used in this project is the ESP32, which controls the entire system. Different sensors like current sensor, voltage sensor, LDR, and IR sensor are connected to it to monitor the system conditions such as voltage level, current flow, light intensity, and vehicle detection. Based on these inputs, the controller automatically turns the charging system ON or OFF using a relay. The system also includes IoT monitoring, where the data is sent to a mobile application so that the user can check the status of charging from anywhere. This project is a simple and cost-effective model that shows how renewable energy and smart monitoring can be used for EV charging.

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CHAPTER - 1

INTRODUCTION

Electric vehicles are becoming an important part of modern transportation due to their environmental benefits and reduced emissions. However, the efficiency of EV usage largely depends on the availability of a reliable and convenient charging system. Most existing charging systems are based on wired connections and rely on grid electricity, which is not sustainable in the long run.One of the major drawbacks of wired charging is energy loss and the inconvenience of handling cables. Also, these systems do not provide real-time monitoring, making it difficult for users to track charging status and system performance.To overcome these issues, this project focuses on developing a smart EV charging system that uses solar energy as a renewable source and wireless power transfer for contactless charging. The system is designed to automatically control charging based on environmental conditions and vehicle presence. IoT integration is also included to enable real-time monitoring and remote access.The main objective of this project is to provide a simple, efficient, and eco-friendly charging solution that reduces dependency on conventional energy sources and improves user convenience.

1.1 Background of the Problem

Electric vehicles are increasing concern over environmental pollution and depletion of fossil fuels, electric vehicles have become an important alternative to conventional vehicles. However, one of the major challenges faced in EV adoption is the availability of efficient and user-friendly charging systems. Most of the existing charging systems rely on wired connections and electricity from the grid. These systems have several disadvantages such as energy loss during transmission, dependency on non-renewable energy sources, and lack of automation. Moreover, the absence of real-time monitoring makes it difficult for users to track the charging status and system performance. Therefore, there is a need for better solution that is efficient, eco-friendly and intelligent

1.2 Motivation and Need for the Project

The motivation behind this project comes from the need to develop a smart charging system that overcomes the limitations of traditional systems. With the increasing demand for renewable energy and automation, it is necessary to design a system that integrates solar power, wireless charging, and IoT monitoring. This project aims to provide a convenient and efficient charging solution by eliminating wires, reducing energy loss, and enabling real-time monitoring. It also promotes the use of clean energy, which helps in reducing environmental impact.

1.3 Problem Statement

With the increasing use of electric vehicles, the demand for efficient and reliable charging systems is also growing. However, most of the existing EV charging systems depend on wired connections and conventional grid electricity. These systems have several drawbacks such as energy loss during transmission, dependency on non-renewable energy sources, and inconvenience due to the use of cables.Another major issue is the lack of automation in charging systems. In many cases, charging has to be manually controlled, which may lead to unnecessary power consumption and inefficient usage of energy. Also, there is limited availability of real-time monitoring, which makes it difficult for users to track the charging status and system performance.In addition, existing systems do not effectively utilize renewable energy sources like solar power. Even if solar energy is used, it is not properly managed or integrated with the charging system. Therefore, there is a need to design a system that can provide wireless charging, use renewable energy efficiently, and allow real-time monitoring with automatic control.

1.4 Objectives of the Project

The main objective of this project is to design and develop a smart electric vehicle charging system that is efficient, eco-friendly, and easy to use. The system aims to reduce dependency on conventional energy sources by using solar energy as the primary source of power. By doing this, the project promotes the use of renewable energy and helps in reducing environmental impact.Another important objective is to implement wireless power transfer for charging the vehicle. This removes the need for physical cables and connectors, making the charging process more convenient and safer for users. It also reduces the wear and tear that usually occurs in wired charging systems.

The project also focuses on monitoring important electrical parameters such as voltage and current using sensors. This helps in understanding the system performance and ensures that the charging process is happening under safe conditions. In addition, an IR sensor is used to detect the presence of a vehicle, which allows the system to operate automatically without manual intervention.One more key objective is to integrate IoT technology into the system. This enables real-time monitoring of the charging process, where the user can check system status such as voltage, current, and charging condition using a mobile device. This improves user convenience and provides better control over the system.

1.5 Scope of the Project

The scope of this project is mainly focused on developing a small-scale prototype of a smart EV charging system that uses solar energy and wireless power transfer. The system is designed for low-power applications such as small electric vehicles or demonstration models. It aims to show how renewable energy and smart technologies can be combined to create an efficient charging system.This project also includes the implementation of IoT-based monitoring, which allows users to observe system parameters such as voltage, current, and charging status in real time. The system is capable of automatic operation based on sensor inputs, which improves efficiency and reduces manual effort.However, the current design is limited in terms of power capacity and wireless charging range. It is not intended for direct use in full-scale EV charging stations. With further improvements, such as increasing power handling capacity and improving wireless efficiency, the system can be extended to real-world applications.

1.6 Organization of the Report

This report is organized into different chapters to provide a clear understanding of the project.

Chapter 1 introduces the basic concept, problem statement, objectives, and scope of the project. It gives an overall idea about the need and importance of the proposed system.

Chapter 2 focuses on the literature survey and existing systems. It discusses previous works, their limitations, and identifies the research gap that led to the development of this project.

Chapter 3 explains the proposed system in detail, including system architecture, working principle, components used, and methodology.

Chapter 4 deals with the design and implementation of the system. It includes circuit design, hardware setup, software development, and system integration.

Chapter 5 presents the results obtained from the project and provides a discussion on system performance.

Chapter 6 concludes the project and discusses possible future improvements and scope for further development.

CHAPTER – 2

LITERATURE AND EXISTING SYSTEM

2.1 Overview of Previous Works

In recent years, a significant amount of research has been carried out in the field of electric vehicle (EV) charging systems. Most of the earlier systems were based on conventional wired charging methods, where vehicles are connected to the power source using cables. These systems are simple and widely used, but they have certain limitations such as energy loss, wear and tear of connectors, and inconvenience to users.

With the advancement in technology, researchers have started focusing on wireless power transfer systems for EV charging. These systems use electromagnetic induction to transfer power from a transmitter coil to a receiver coil without physical contact. This approach improves safety and convenience, as it eliminates the need for cables. However, many of these systems are still in the experimental stage and are not widely implemented due to cost and efficiency issues.

At the same time, renewable energy-based charging systems have also gained attention. Solar-powered charging stations are being developed to reduce dependency on conventional electricity and to promote sustainable energy usage. These systems convert sunlight into electrical energy using solar panels and store it in batteries for later use. However, many existing solar charging systems do not include automation or smart monitoring features.

In addition, IoT-based monitoring systems have been introduced in recent works to track system performance. These systems allow users to monitor parameters such as voltage, current, and charging status remotely. Despite these advancements, most of the existing works focus on individual features like wireless charging, solar power, or IoT, rather than integrating all of them into a single system.

2.2 Limitations in Existing Systems

Even though electric vehicle charging systems are widely used, they still have several limitations which affect their efficiency and usability. One of the main limitations is the dependence on conventional grid electricity. Most of the existing charging systems rely on power generated from non-renewable sources, which increases environmental pollution and is not sustainable in the long run.Another major limitation is the use of wired charging. In these systems, physical cables are required to connect the vehicle to the charging station. This can be inconvenient for users and may also lead to wear and tear of connectors over time. In some cases, improper connections can also cause safety issues.Energy loss is also a significant problem in wired charging systems. During transmission through cables, some amount of energy is lost in the form of heat, which reduces overall efficiency. Additionally, these systems do not have proper automation, meaning that the charging process often requires manual control. This can result in unnecessary power consumption if the system is not managed properly.Most existing systems also lack real-time monitoring features. Users cannot easily track important parameters such as voltage, current, and charging status. This makes it difficult to understand system performance and detect faults at an early stage.Furthermore, many systems do not effectively utilize renewable energy sources like solar power. Even if solar energy is used, it is often not integrated properly with the charging system. Because of these limitations, there is a need for a more advanced system that is efficient, automated, and capable of real-time monitoring.

2.3 Research Gap Identified

From the study of existing systems and previous research works, it is clear that there are still several gaps that need to be addressed. Most of the currently available EV charging systems rely on wired connections and grid electricity, which makes them less efficient and not environmentally friendly. Even though wireless charging systems are being developed, they are not widely adopted due to limitations in efficiency and implementation complexity.Similarly, solar-based charging systems are available, but many of them lack proper automation and control mechanisms. They do not efficiently manage the available energy or adapt to changing conditions such as variations in sunlight or battery levels. This results in inefficient utilization of resources. Another important gap is the lack of real-time monitoring and integration. While IoT technology is used in some systems, it is often implemented separately and not fully integrated with the charging process. There is a need for a system where monitoring, control, and charging work together seamlessly. Therefore, there is a clear requirement for a system that combines solar energy, wireless power transfer, and IoT-based monitoring into a single, efficient solution. The proposed project aims to address these gaps by developing a system that is not only eco-friendly but also smart, automated, and user-friendly

2.4 Summary

From studying the existing EV charging systems, it is understood that most of them are based on wired charging and depend on electricity from the grid. Even though these systems are commonly used, they have some problems like energy loss, inconvenience due to cables, and no proper flexibility. Also, they do not make effective use of renewable energy sources like solar power, which is very important nowadays.There are some improvements in recent technologies like wireless charging, solar-based systems, and IoT monitoring. Wireless charging makes the process easier by removing wires, and solar energy helps in reducing dependency on normal electricity. IoT is useful for checking system status in real time. But in most cases, these features are used separately and not combined into a single system.Another thing observed is that many systems do not have proper automation. They do not respond automatically to conditions like sunlight availability, battery level, or vehicle presence. Because of this, energy is not used efficiently and overall performance is reduced.So, there is a need for a system that combines all these features together in a simple and effective way. The proposed project focuses on solving these problems by using solar energy, wireless charging, and IoT monitoring in one system. This helps in making the system more efficient, eco-friendly, and easy to use.

CHAPTER – 3

PROPOSED SYSTEM / METHODOLOGY

3.1 System Architecture / Block Diagram

The system architecture of the Smart Solar Wireless EV Charging Station consists of several interconnected components that work together to perform charging and monitoring functions. The main blocks in the system include the solar panel, battery, voltage regulating unit, controller, sensors, relay module, and wireless charging module.Initially, the solar panel acts as the primary energy source. It converts sunlight into electrical energy using the photovoltaic effect. This generated energy is then supplied to the battery, where it is stored for later use.The battery plays an important role as it provides continuous power to the system even when sunlight is not available.Since the output voltage of the battery may not be constant, a voltage regulating unit is used.

In this project, a boost converter is used to increase and stabilize the voltage to the required level (usually 5V). This regulated voltage is then supplied to the controller and other components to ensure proper functioning.The main controller used in the system is the ESP32, which acts as the brain of the entire system. It receives input from various sensors and controls the operation of the system based on those inputs.Different sensors are connected to the controller to monitor system conditions. The voltage sensor measures the battery voltage, and the current sensor measures the current flowing in the circuit.

The LDR sensor is used to detect the intensity of sunlight, which helps in understanding whether solar energy is available. The IR sensor is used to detect the presence of a vehicle near the charging station.The relay module is connected between the battery and the wireless transmitter. It acts as a switching device and controls the flow of power. When the controller sends a signal, the relay turns ON and allows power to flow to the transmitter. Otherwise, it remains OFF and stops the power supply.

The wireless charging module consists of a transmitter coil and a receiver coil. The transmitter coil generates a magnetic field when powered, and the receiver coil captures this energy and converts it back into electrical energy to charge the vehicle.In addition to these components, the system also includes IoT functionality. The controller uses WiFi to send sensor data to a cloud platform, enabling real-time monitoring through a mobile device.Overall, the system architecture is designed in such a way that all components are interconnected and work together efficiently to provide automatic, wireless, and solar-based EV charging with real-time monitoring.

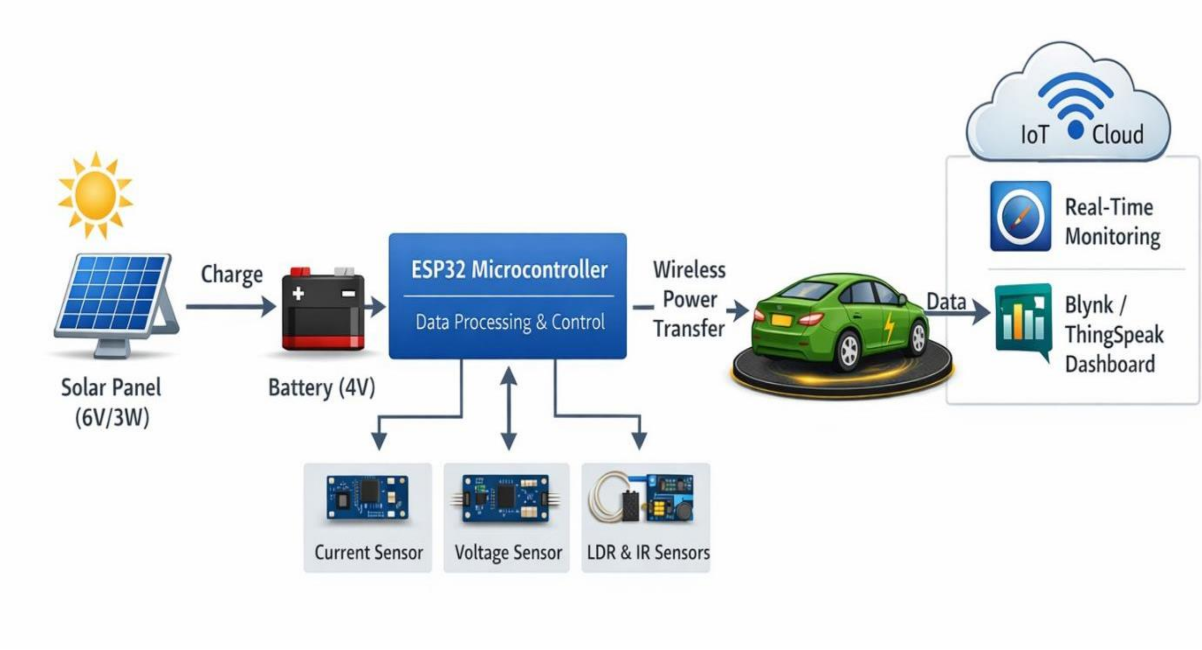


Fig. 3.1 Block Diagram

3.2 Working Principle

Initially, the solar panel converts sunlight into electrical energy through the photovoltaic effect. This generated energy is used to charge the battery, which acts as the main power source for the system. The battery stores energy so that charging can take place even when sunlight is not available.The stored energy from the battery is supplied to the system through a voltage regulating unit. Since the battery voltage may not be constant, a boost converter is used to increase and maintain a stable output voltage (typically 5V) required for the wireless charging module and controller.The ESP32 acts as the central controller of the system. It continuously reads data from different sensors connected to it. The voltage sensor measures the battery voltage, and the current sensor measures the current flowing in the circuit.

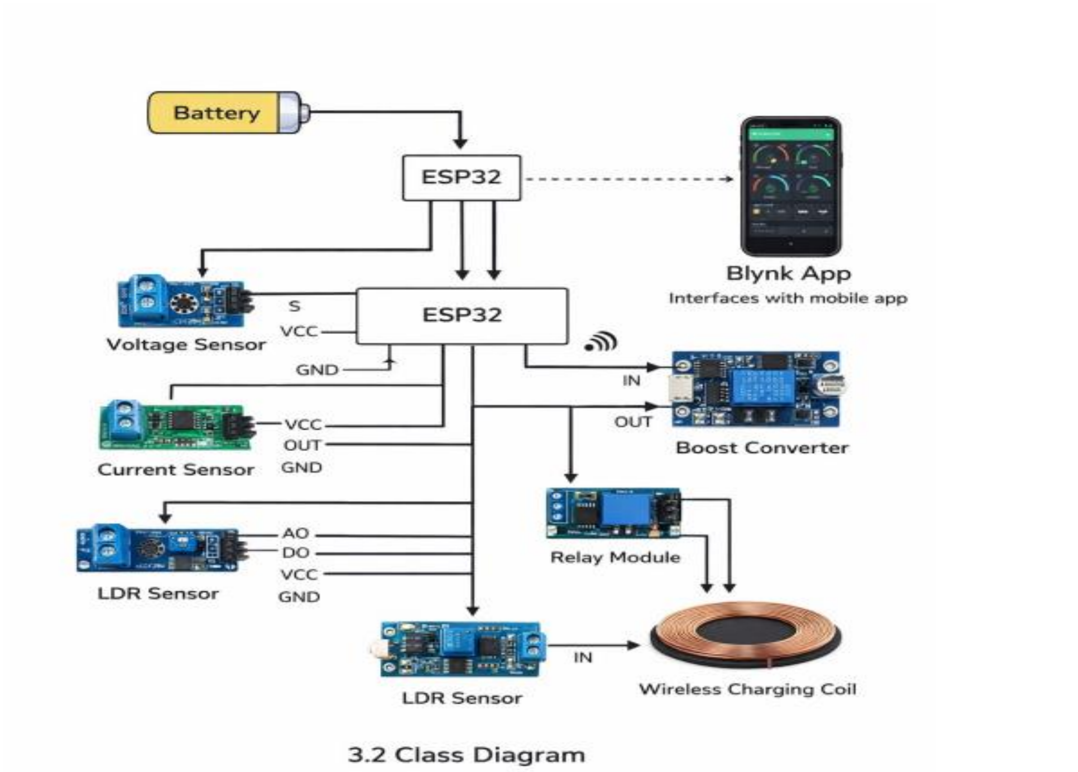
The LDR sensor detects the availability of sunlight, and the IR sensor detects the presence of a vehicle near the charging station.When the system is active, the controller checks all the conditions. If sufficient battery voltage is available and a vehicle is detected, the controller sends a signal to activate the relay module. The relay acts as a switch and allows power to flow from the battery to the wireless transmitter.The wireless charging module works on the principle of electromagnetic induction. The transmitter coil generates an alternating magnetic field when powered. This magnetic field is received by the receiver coil placed in the vehicle.

The receiver converts this magnetic energy back into electrical energy, which is then used to charge the vehicle battery.At the same time, the ESP32 sends all the sensor data such as voltage, current, and charging status to an IoT platform using WiFi. This allows the user to monitor the system in real time through a mobile application or web interface.

If any condition is not satisfied, such as low battery voltage or absence of a vehicle, the controller automatically turns OFF the relay, stopping the charging process. This ensures efficient energy usage and protects the system from damage.Overall, the system works automatically without manual intervention, combining renewable energy, wireless power transfer, and smart monitoringto provide an efficient EV charging solutions.

3.3 Detailed Explanation of Each Block/Component

ESP32: The main controller used in this project is the ESP32, which acts as the brain of the system. It is responsible for controlling all operations and making decisions based on sensor



inputs. The ESP32 reads data from different sensors such as voltage sensor, current sensor, LDR, and IR sensor. Based on these values, it controls the relay module to start or stop the charging process.In addition, the ESP32 has built-in WiFi capability, which allows it to connect to the internet and send data to an IoT platform. This helps in real-time monitoring of system parameters. The controller continuously runs the program and ensures automatic operation without manual intervention.

Current Sensor: The current sensor is used to measure the current flowing through the circuit. It is usually connected in series with the load. This sensor helps in monitoring how much current is being used during the charging process.

Voltage Sensor: The voltage sensor is used to measure the voltage level of the battery. It helps the controller to monitor whether the battery has sufficient charge. The sensor works by reducing the input voltage to a safe level that can be read by the ESP32.

LDR Sensor: The Light Dependent Resistor is used to detect the intensity of sunlight. Its resistance changes based on the amount of light falling on it.When sunlight is available, the LDR gives a signal indicating that solar energy can be used. This helps the system to operate efficiently by using available solar power.

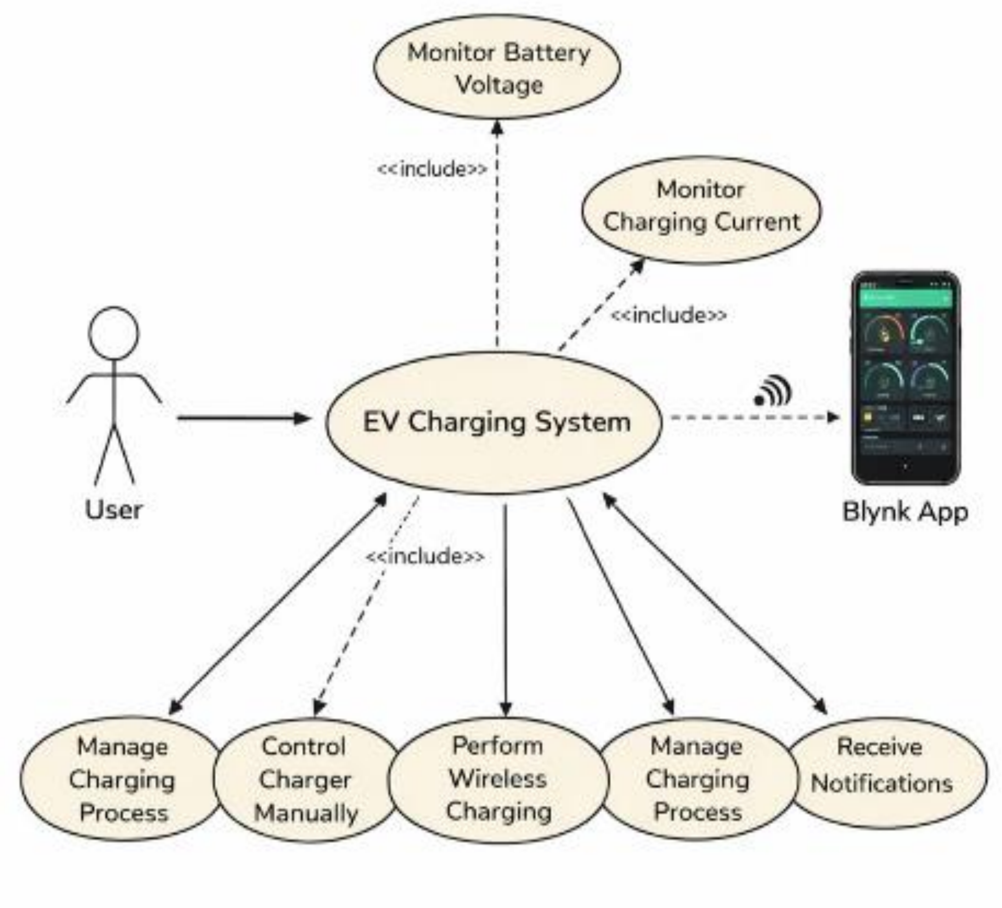
IR Sensor: The Infrared Obstacle Avoidance Sensor Module is used to detect the presence of a vehicle near the charging station. It works by emitting infrared rays and detecting their reflection.When a vehicle is detected, the sensor sends a signal to the controller. This allows the system to automatically start charging without manual operation.

Relay Module: The Relay Module acts as a switching device in the system. It is controlled by the ESP32 and is used to turn the charging system ON or OFF.When the controller sends a signal, the relay closes the circuit and allows power to flow to the wireless transmitter. When the signal is OFF, the relay opens the circuit and stops the power supply.

Wireless Charging Module: The Wireless Power Transfer Module is used to transfer power without wires. It consists of a transmitter coil and a receiver coil.When power is supplied to the transmitter, it generates a magnetic field. This magnetic field is received by the receiver coil, which converts it back into electrical energy to charge the vehicle battery. This method provides a convenient and safe charging solution.

Booster Converter: The DC-DC Boost Converter is used to increase the battery voltage to a required stable level. Since the battery voltage may be lower than the required voltage for the components, the boost converter steps up the voltage (for example, from around 4.3V to 5V).It provides a constant and regulated output, which is necessary for proper functioning of the ESP32 and wireless charging module. Without this, voltage fluctuations can damage components or affect performance.

3.3 UsecaseDiagram



3.3 Usecase Diagram

The use case diagram represents how the user interacts with the EV charging system and how different functions of the system are performed. It shows the relationship between the user, the system, and various operations involved in the charging process.In this diagram, the main actor is the **user**, who interacts with the EV Charging System. The system acts as the central unit that performs all operations such as monitoring, controlling, and charging.

Main System: EV Charging System

The EV Charging System is the core part of the diagram. It includes all functionalities like monitoring battery voltage, measuring charging current, managing the charging process, and performing wireless charging. All actions are controlled by the system based on inputs from sensors and user interaction.

Monitoring Functions:

1. **Monitor Battery Voltage:**This function is used to continuously check the battery voltage level. It helps in ensuring that the battery has enough charge to perform charging. If the voltage is low, the system may stop charging to protect the battery.
2. **Monitor Charging Current:**This function measures the current flowing during the charging process. It helps in understanding how much power is being transferred and ensures safe

operation by avoiding overload conditions.

3. **Charging Control Functions**

4. **Manage Charging Process:**This represents the overall control of the charging operation. It decides when to start or stop charging based on conditions like battery level, vehicle detection, and system status.

5. **Control Charger Manually:**This function allows the user to manually control the charging system. The user can turn the charger ON or OFF using the mobile application or interface.

6. **Perform Wireless Charging:**This is the main function where power is transferred wirelessly from the transmitter to the receiver coil. It happens only when all conditions are satisfied, such as vehicle detection and sufficient power availability.

User Interaction with IoT:

Blynk App:

The system is connected to the Blynk, which allows the user to monitor and control the system remotely.

Through this app, the user can:

- Check battery voltage
- Monitor charging current
- Start or stop charging
- Receive updates about system status

Receive Notifications:

This function provides alerts to the user. Notifications can be sent when:

- Charging starts or stops
- Battery is low
- Any fault occurs

This improves user awareness and system safety.

3.4 Flowchart or Algorithm Steps:

The flowchart represents the working process of the smart solar-based EV charging system. It shows how the system operates step by step and makes decisions based on different conditions.

Start:

The process begins when the system is powered ON. This means the battery or solar panel starts supplying power to the system.

Initialize System: After starting, the system initializes all components. The ESP32 sets up all sensors, relay module, and IoT connection. This step ensures that every part of the system is ready for operation.

Read Sensor Values:

Next, the system reads input values from all sensors:

· Voltage sensor checks battery level

· Current sensor measures current

· LDR sensor checks sunlight intensity

· IR sensor detects vehicle presence

This step is repeated continuously to monitor real-time conditions.

Check Battery and Sunlight:

The system checks whether:

· The battery has enough charge

· Solar energy (sunlight) is available

If these conditions are not satisfied, the system will not proceed further for charging.

Detect Vehicle:

The IR sensor checks if a vehicle is present near the charging station.

· If no vehicle is detected → charging will not start

· If a vehicle is detected → system moves to the next step

Condition Decision:

At this stage, the system verifies all conditions together:

Battery level is sufficient

Sunlight is available

Vehicle is present

If the conditions are **NOT satisfied**:

- Relay is turned OFF
- Charging is stopped

If the conditions are **satisfied**:

- Relay is turned ON
- Charging process begins.

Relay Operation:

The relay acts as a switch:

Relay OFF → No power supply → No charging

- **Relay ON** → Power supply given → Charging starts

Charging Process:

When the relay is ON:

- Power flows to the wireless charging system
- Energy is transferred to the vehicle
- Charging takes place automatically

Send Data to IoT:

After each operation, the system sends data such as voltage, current, and charging status to the mobile application using IoT. This allows the user to monitor the system in real time through the Blynk app.

Repeat Process:

The system continuously repeats the same steps:

· Read sensors

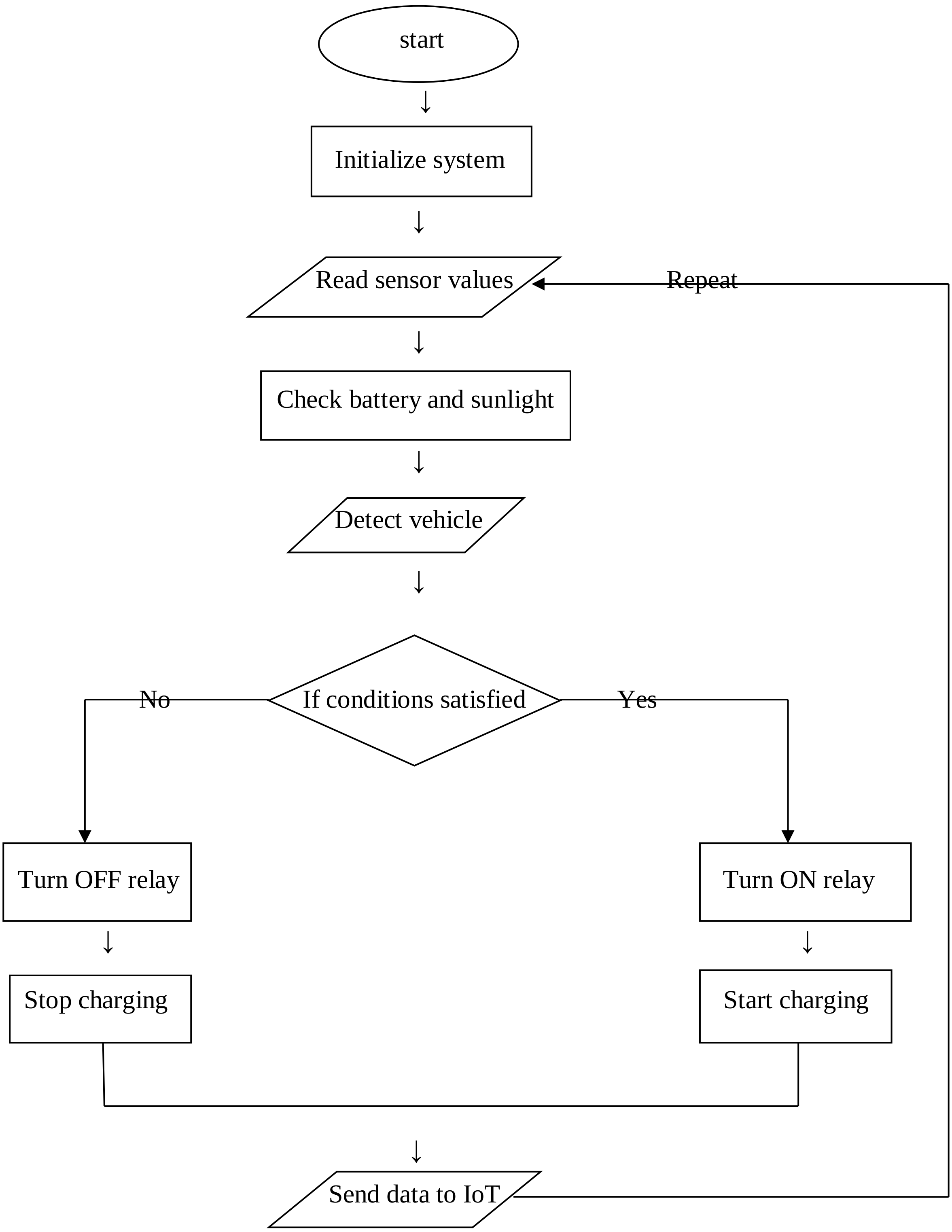
· Check conditions

· Control charging

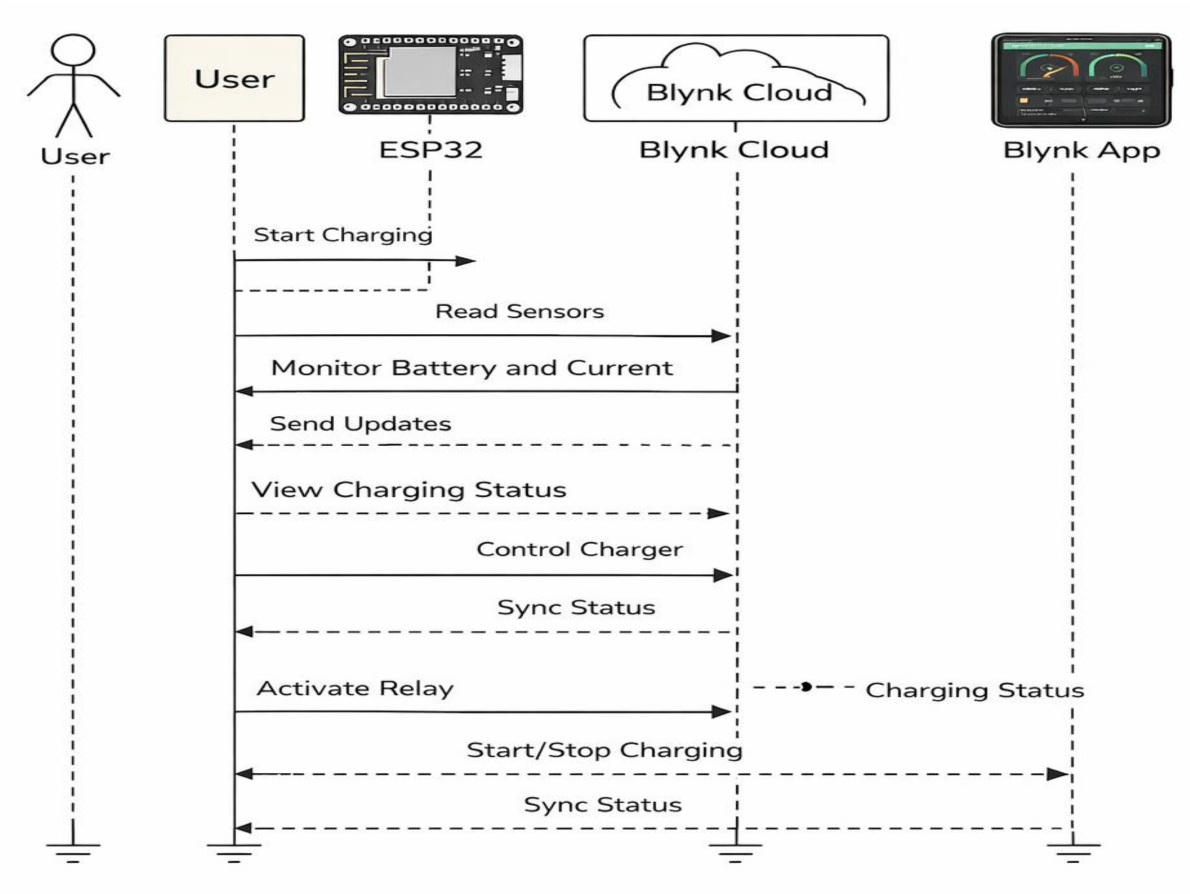
This loop ensures automatic and real-time operation.

The flowchart clearly shows that the system works automatically by checking battery, sunlight, and vehicle presence.Based on these conditions, it controls the charging process and sends real-time updates to the user.

FLOW CHART



Flow Chart



3.4 Sequence Diagram

User Initiation:

The process begins when the user starts the charging operation through the mobile application.This action sends a request to the ESP32, which acts as the main controller of the system and handles all further operations.

Sensor Data Collection:

After receiving the command, the ESP32 starts reading data from all connected sensors.These include voltage, current, sunlight intensity, and vehicle detection sensors.This step helps the system understand the current operating conditions before proceeding charging.

Monitoring and Analysis:

The ESP32 then analyzes the collected sensor data to check whether the battery level is sufficient and whether the charging conditions are safe.This step ensures that the system avoids issues like over charging.

Cloud Communication:

Once the data is processed, it is sent to the Blynk Cloud. The cloud acts as an intermediate platform that stores and transfers data between the hardware and the mobile application.

User Monitoring:

The user can view all the system parameters through the Blynk app.Information such as voltage, current, and charging status is displayed in real time,allowing the user to monitor system easily.

Control of Charging:

The user also has the ability to control the charger manually.Commands like turning the charging ON or OFF are sent from the mobile app to the cloud,and then to the esp32 for execution.

Relay Operation:

Based on the received commands and system conditions, the ESP32 activates or deactivates the relay. When the relay is ON, power is supplied to the charging system and charging begins. When the relay is OFF, the charging process stops.

Charging Process:

When all conditions are satisfied and the relay is activated, the system starts charging the vehicle.If conditions are not met or the user stops the process,charging is immediately halted.

Status Synchronization:

Throughout the operation, the system continuously synchronizes data between the ESP32, cloud, and mobile application.This ensures that all components display the same real time information.

Finally, the charging status and system data are updated and shown to the user through the mobile application. This provides continuous feedback and ensures proper monitoring of the entire charging process.

3.5 Hardware and Software Requirements

Hardware:

1. Solar panel
2. Battery
3. ESP32
4. Voltage Sensor
5. Current Sensor
6. LDR Sensor
7. IR Sensor
8. Relay module
9. wireless charging module
10. DC-DC booster converter

Software:

1. Arduino IDE
2. Blynk IoT platform for monitoring

Activity Diagram Explanation:

Start and Initialization:

The activity begins when the system is powered ON and the charging process is initiated.At this stage, the ESP32 initializes all components and prepares the system for operation. It sets up sensors, relay module, and communication with the IoT platform to ensure smooth functioning.

Reading Sensor Data:

Once initialized, the system starts reading data from all connected sensors. These include voltage, current, and vehicle detection sensors.This step is very important because the system depends on real-time data to make decisions about charging.

Manual Control Decision:

After collecting sensor data, the system checks whether manual control is enabled through the Blynk app. If manual control is active, the user can directly start or stop the charging process.If the user selects to stop charging, the system immediately deactivates the relay,cutting off power supply and stopping the charging operation.

Automatic Mode Operation:

If manual control is not enabled, the system continues in automatic mode. In this mode, the ESP32 sends sensor data to the Blynk Cloud.The cloud then updates the mobile app, allowing the user to monitor the system in real time without manual intervention.

Data Display and Monitoring:

The collected data such as voltage, current, and charging status is displayed in the mobile application.This helps the user to understand the system performance and monitor charging conditions continuously.

Relay Activation Decision:

The system then checks whether to activate the relay. If all required conditions are satisfied, such as vehicle presence and sufficient battery level,the relay is turned ON, and charging starts. If conditions are not met, the relay remains OFF, preventing charging.

Vehicle Detection and Battery Check

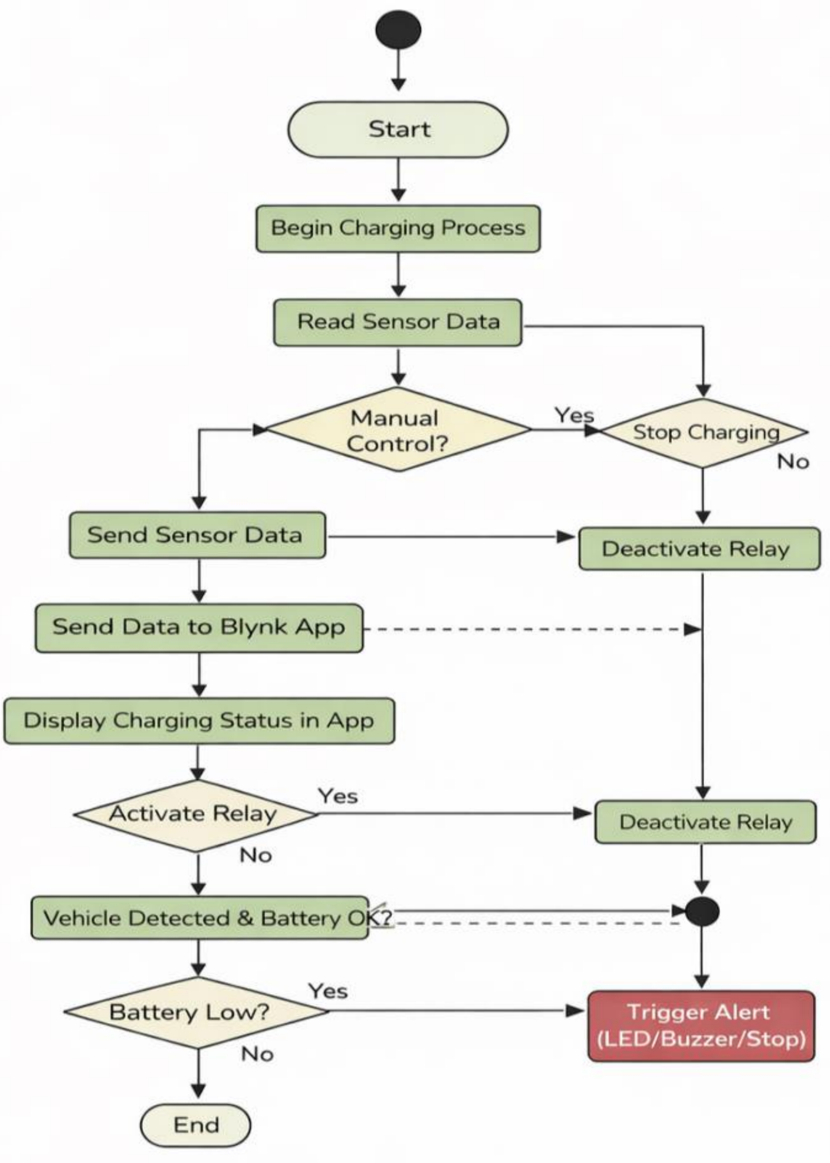
Next, the system verifies whether a vehicle is detected and whether the battery condition is suitable for charging. If both conditions are satisfied, the system continues the charging process. Otherwise, charging is not allowed to ensure safety.

Battery Low Condition

If the system detects that the battery level is low, it triggers an alert. This alert can be in the form of an LED indication, buzzer sound, or automatic stopping of the charging process. This protects the system from damage and ensures safe operation.

End Process:

If all conditions are normal and no issues are detected, the process continues smoothly until completion and then reaches the end state. The system keeps monitoring continuously to maintain proper operation.



3.5 Activity Diagram

CHAPTER – 4

DESIGN AND IMPLEMENTATION

4.1 Circuit Diagrams / PCB Layouts

The circuit of the proposed system is designed by connecting all components such as the solar panel, battery, controller, sensors, relay module, and wireless charging module in a proper sequence.Initially, the solar panel is connected to the battery to store the generated energy. The battery acts as the main power source for the entire system. Since the battery voltage is not always stable, a DC-DC boost converter is used to provide a regulated output voltage required for the controller.The ESP32 is powered through the boost converter. All the sensors are connected to the input pins of the controller. The current sensor is connected in series with the load to measure current flow, while the voltage sensor is connected across the battery to measure voltage.The LDR sensor is connected using a voltage divider circuit to detect light intensity, and the IR sensor is connected to detect the presence of a vehicle. The relay module is connected between the battery and the wireless transmitter to control the charging process.The overall circuit is initially implemented on a breadboard for testing purposes. Proper connections are ensured to avoid short circuits and ensure stable operation.

4.2 Hardware Design

The hardware design of the system includes all physical components required to implement the project. The main components used are solar panel, battery, controller, sensors, relay, and wireless charging module.The solar panel is used to convert sunlight into electrical energy. This energy is stored in a rechargeable battery, which acts as a backup power source. The battery supplies power to the controller and other components.The central controller used is the ESP32, which is responsible for controlling the entire system. It is connected to different sensors to monitor system conditions.The current sensor and voltage sensor are used to measure electrical parameters.

The LDR sensor detects the presence of sunlight, which helps in deciding whether solar energy is available. The IR sensor is used to detect whether a vehicle is present in the charging area.A relay module is used as a switching device to control the power supply to the wireless charging module. The wireless charging system consists of a transmitter and receiver coil. The transmitter is connected to the power supply, and the receiver is placed in the vehicle.All components are connected using wires and placed properly to ensure safety and proper functioning. The hardware setup is compact and suitable for demonstration.

4.3 Software Design

The main function of the software is to read data from sensors, process the data, and control the relay accordingly. The program continuously reads the values from the current sensor, voltage sensor, LDR, and IR sensor.Based on the sensor inputs, the controller decides whether the charging should be turned ON or OFF. For example, if the battery has sufficient charge, sunlight is available, and a vehicle is detected, the controller activates the relay and starts charging.If any of these conditions are not satisfied, the relay is turned OFF to stop charging. This ensures efficient utilization of energy and prevents unnecessary power consumption.

```
#define BLYNK_TEMPLATE_ID "TMPL3HhbF-mKf"
#define BLYNK_TEMPLATE_NAME "EV Charging System"
#define BLYNK_AUTH_TOKEN "Qq0d8xjR0St5C0NEujQoSYiPkzbGPnE6"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

char ssid[] = "Monster";
char pass[] = "11112222";
BlynkTimer timer;

// ---------------- PIN CONFIGURATION ----------------

#define RELAY_PIN 23
#define BUZZER_PIN 25

#define GREEN_LED 14
#define RED_LED 27
#define YELLOW_LED 26

#define IR_PIN 18
#define LDR_PIN 32
#define VOLTAGE_PIN 34
#define CURRENT_PIN 35

// ---------------- VARIABLES ----------------

float voltage = 0;
float current = 0;
```

```
float power = 0;
int ldrValue = 0;
bool vehicleDetected = false;
bool manualControl = false;

// ---------------- SENSOR FUNCTIONS----------------

void readSensors()
{
  int v Raw = analog Read(VOLTAGE_PIN);
  int c Raw = analog Read(CURRENT_PIN);

  voltage = (vRaw * 3.3 / 4095.0) * 4.0;   // adjust factor based on divider
  current = (cRaw * 3.3 / 4095.0) * 2.0;   // calibration required

  power = voltage * current;

ldrValue = analogRead(LDR_PIN);
Vehicle Detected = (digitalRead(IR_PIN) == LOW);

  // Send to Blynk
Blynk.virtualWrite(V0, voltage);
Blynk.virtualWrite(V1, current);
Blynk.virtualWrite(V2, power);
Blynk.virtualWrite(V3, ldrValue);
Blynk.virtualWrite(V4, vehicleDetected);
}

// ---------------- AI LOGIC ----------------

void smartControl()
{
  if(!manualControl)
  {
if(vehicleDetected&& voltage > 3.5)
```

```
    {
digitalWrite(RELAY_PIN, HIGH);  // ON
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);
    }
    else
    {
digitalWrite(RELAY_PIN, LOW);   // OFF
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
    }

    // Low battery protection
if(voltage < 3.5)
    {
digitalWrite(YELLOW_LED, HIGH);
digitalWrite(BUZZER_PIN, HIGH);
    }
    else
    {
digitalWrite(YELLOW_LED, LOW);
digitalWrite(BUZZER_PIN, LOW);
    }
  }
}

// ---------------- MANUAL CONTROL ----------------

BLYNK_WRITE(V5)
{
  int value = param.asInt();
manualControl = true;

digitalWrite(RELAY_PIN, value);
```

```
if(value == 1)
  {
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);
  }
  else
  {
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
  }
}

// ---------------- SETUP ----------------

void setup()
{
Serial.begin(9600);

pinMode(RELAY_PIN, OUTPUT);
pinMode(BUZZER_PIN, OUTPUT);

pinMode(GREEN_LED, OUTPUT);
pinMode(RED_LED, OUTPUT);
pinMode(YELLOW_LED, OUTPUT);

pinMode(IR_PIN, INPUT);

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

timer.setInterval(1000L, []() {
readSensors();
smartControl();
  });
}
```

```
// ---------------- LOOP ----------------

void loop()
{
Blynk.run();
timer.run();
}
```

4.4 Communication Protocols Used

The communication in this system is mainly based on WiFi technology. The controller used in this project has built-in WiFi capability, which allows it to connect to the internet.UsingWiFi, the system sends data such as voltage, current, and charging status to an IoT platform. This data can be accessed remotely through a mobile application or web interface.The communication is continuous, and data is updated in real time. This helps the user to monitor the system performance and charging status from anywhere. WiFi communication is chosen because it is easy to implement, cost-effective, and widely available.

4.5 System Integration Details

System integration is the process of combining all hardware and software components into a single working system. In this project, all components are connected and tested step by step. First, the power supply section is verified by checking the solar panel and battery connections. Then, the controller is powered and tested. After that, sensors are connected and tested individually to ensure correct readings. Next, the relay module is connected and tested to verify switching operation. Finally, the wireless charging module is connected, and the complete system is tested. The software is then integrated with the hardware, and the entire system is tested under different conditions. The IoT platform is also connected, and data transmission is verified. Proper integration ensures that all components work together smoothly and efficiently. The final system operates automatically, providing charging and monitoring without manual intervention.

CHAPTER – 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

The experimental setup of the project was carried out by assembling all the required components in a step-by-step manner. Initially, the solar panel was connected to the rechargeable battery to store the generated energy. The battery was then used as the main power source for the system. A boost converter was connected to regulate the voltage and provide a stable output required for proper functioning of the components.

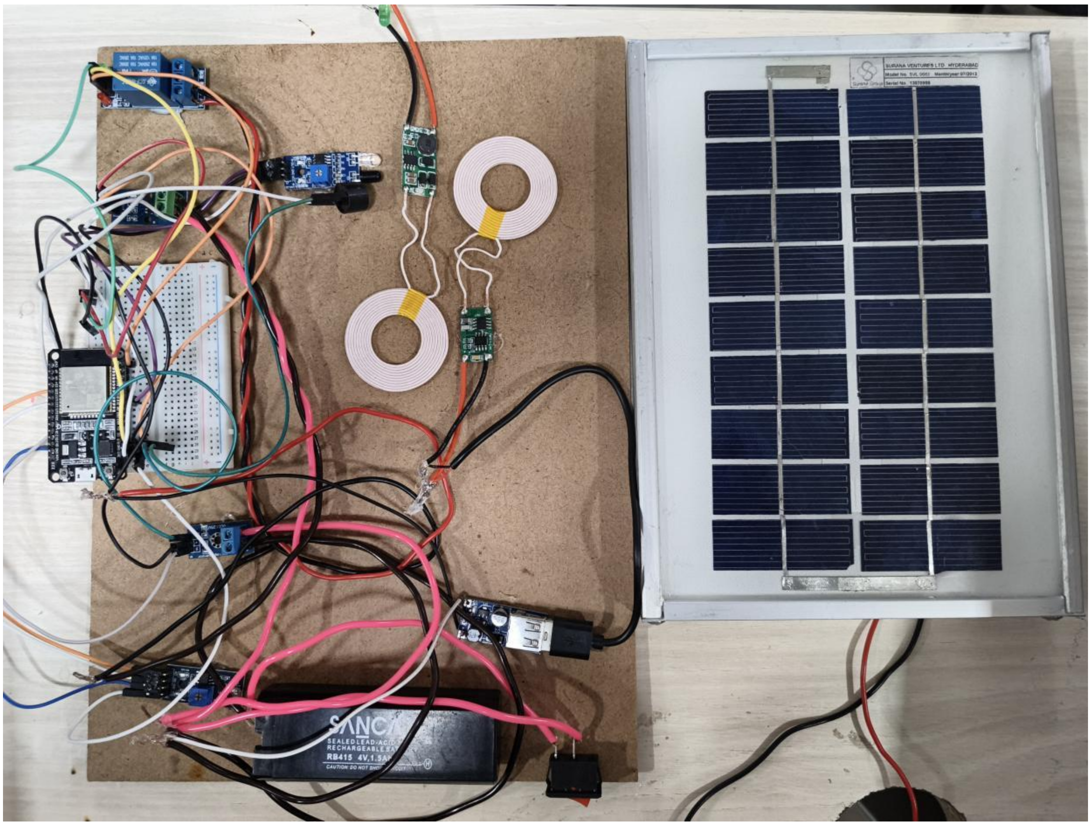
The ESP32 was used as the central controller and was powered using the regulated supply. All sensors such as the voltage sensor, current sensor, LDR sensor, and IR sensor were connected to the controller. These sensors were carefully tested individually to ensure that they were giving correct readings.The relay module was connected between the power source and the wireless transmitter to control the charging process. The wireless charging module, consisting of transmitter and receiver coils, was placed at an appropriate distance to ensure proper power transfer. The entire system was arranged on a breadboard and tested under different conditions such as varying sunlight, battery levels, and vehicle presence.

5.2 Output Results

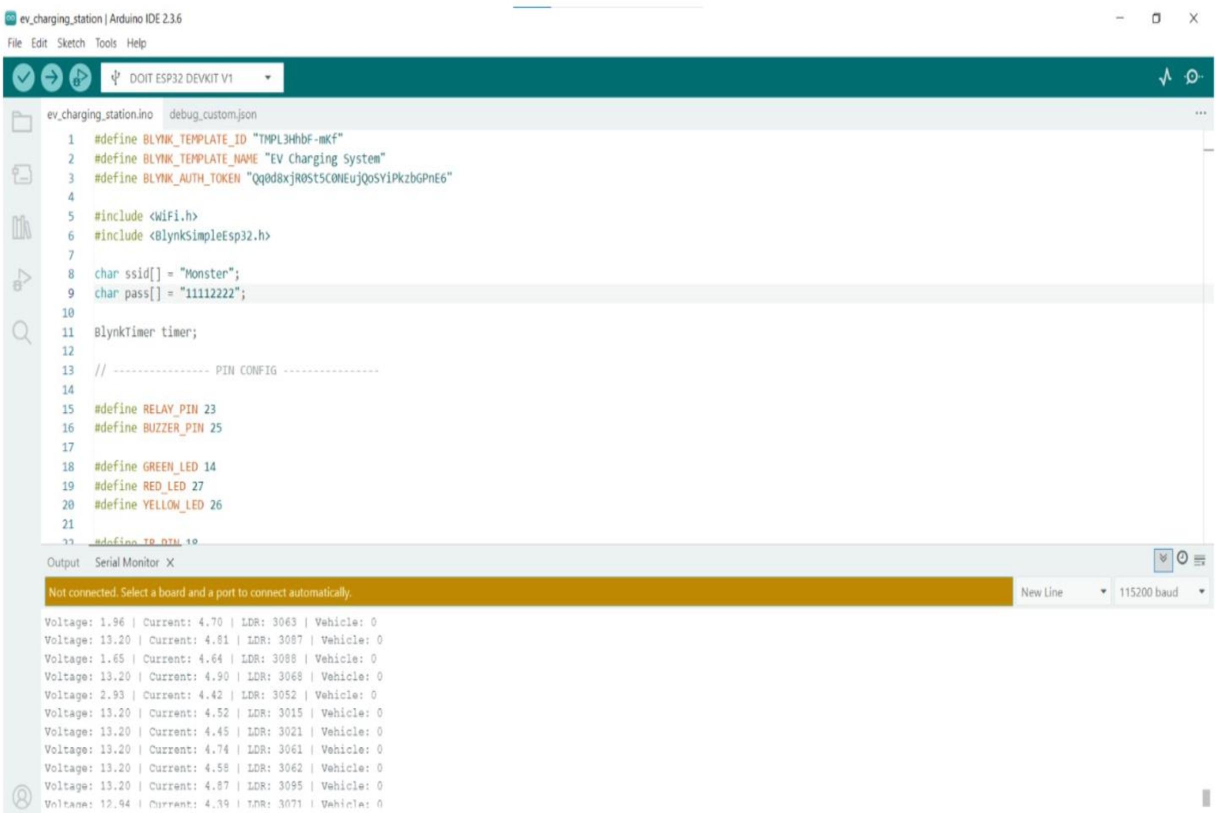
The system produced expected results during testing. When the solar panel was exposed to sunlight, it successfully generated electrical energy, which was stored in the battery. The boost converter provided a stable voltage output, ensuring smooth operation of the controller and other components.

When a vehicle was detected using the IR sensor and sufficient battery voltage was available, the relay module was activated. This allowed power to flow to the wireless transmitter, which then transferred energy to the receiver coil through electromagnetic induction. The charging process was successfully initiated without any physical connection.

The sensors provided real-time data, and the controller processed this information accurately. The IoT feature enabled the transmission of data such as voltage, current, and charging status to a mobile device. The system responded correctly by turning OFF the relay when conditions were not satisfied, such as low battery or absence of a vehicle.



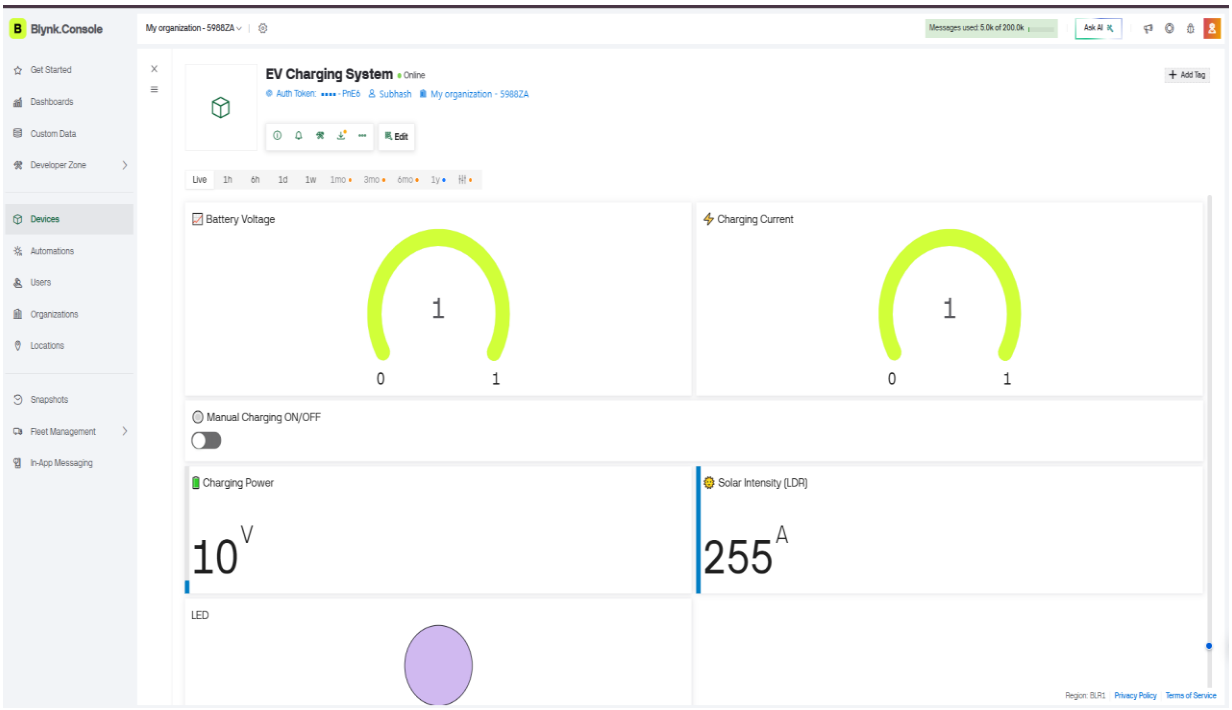
Experimental Setup



Output results in Arduino IDE

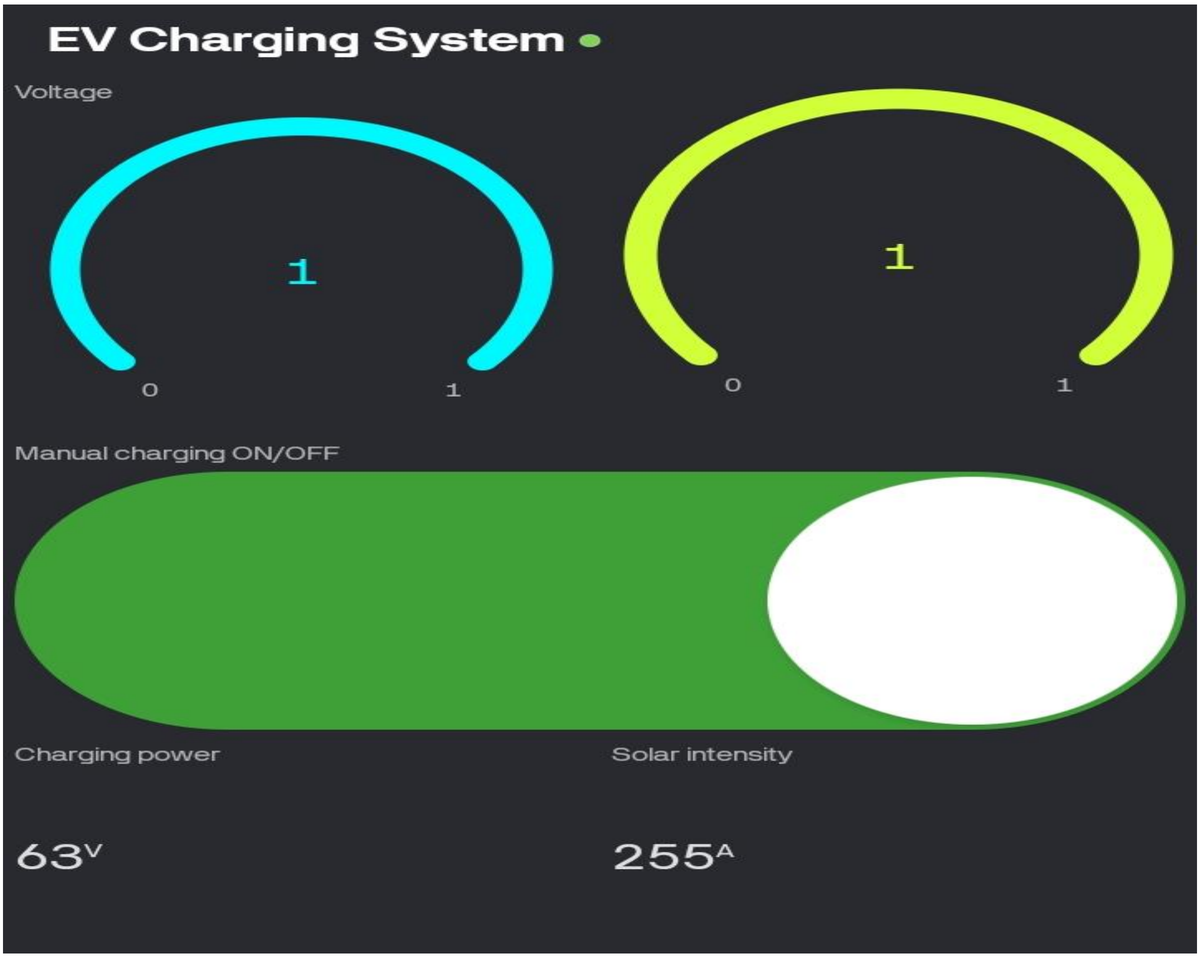
5.3 Performance Analysis

The overall performance of the system was found to be stable and reliable for small-scale applications. The use of solar energy proved to be effective under proper sunlight conditions, and the battery provided backup power when sunlight was not available. The boost converter ensured that a constant voltage was maintained, which improved system efficiency.The wireless charging system worked efficiently within a limited distance, showing good energy transfer capability for low-power applications. The use of sensors helped in automating the system, reducing manual effort and preventing unnecessary energy loss. The integration of IoT allowed continuous monitoring, which enhanced the usability and control of the system.However, some limitations were observed, such as reduced efficiency in low sunlight conditions and limited range of wireless power transfer. Despite these limitations, the system demonstrated a successful implementation of a smart, eco-friendly, and automated EV charging solution.



Blynk web dashboard

Interface on blynk app



Interface on blynk app

Voltage Monitoring:

In the first screen, the voltage gauge shows a value close to 1V, and in the second screen it shows around 7V.

This indicates that the voltage sensor is successfully measuring the battery or charging voltage and sending the data to the mobile application.

A **low voltage (1V)** may represent:

· Low battery condition

· Initial testing stage

A **higher voltage (7V)** shows:

· Proper battery charging

· Stable system operation

3. **Current Monitoring:**

The current gauge in both screens shows a value of 255A (maximum scale).

· The sensor is giving maximum analog value (0–255 rang**e)**

· It may not be actual current but a scaled or uncalibrated value

· The current sensor is sending data, but

· It may need calibration for accurate real-world readings.

4. **Manual Charging Control:**

The ON/OFF switch represents manual control of the charging system.

In the first image, the switch is ON (green)

Charging system is active

Relay is turned ON

In the second image, the switch is OFF (grey)

Charging system is stopped

Relay is turned OFF

5. **Charging Power Display:**

The charging power value is shown below the switch.

· In the first image: around 63V (or calculated value)

· In the second image: around 7V

This value is derived from voltage and current readings. It indicates how much power is being delivered to the load.

- System responds to changing conditions
- Power depends on battery and solar input

5. Solar Intensity

The solar intensity is shown as 255A (maximum value) in both images.

This indicates:

- LDR sensor is detecting high light intensity
- Or giving maximum analog output

This means:

- Solar energy is available
- System can use solar power for charging

Overall System Behavior

From the results, we can observe:

- Sensors are successfully collecting data
- ESP32 is transmitting data through WiFi
- Blynk app is displaying real-time values
- User can control charging remotely
- System responds to ON/OFF commands correctly

CHAPTER – 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

In this project, a Smart Solar Wireless EV Charging Station with IoT Real-Time Monitoring was successfully designed and implemented. The system makes use of solar energy as a renewable power source, which helps in reducing dependency on conventional electricity and promotes eco-friendly operation. The use of a rechargeable battery ensures that the generated energy can be stored and used whenever required.The wireless power transfer technique used in this project eliminates the need for physical connections, making the charging process more convenient and safe. The integration of the ESP32 as the main controller enables efficient monitoring and control of the system.

Sensors such as current sensor, voltage sensor, LDR, and IR sensor help in analyzing system conditions and allow automatic control of the charging process through a relay module.The addition of IoT monitoring provides a significant advantage by allowing users to observe system performance in real time. Parameters such as voltage, current, and charging status can be viewed remotely using a mobile device. This improves user awareness and system management.Overall, the project demonstrates a simple, cost-effective, and efficient solution for EV charging using renewable energy and smart monitoring techniques. Although the current system is designed for low-power applications, it clearly shows the feasibility of implementing such systems on a larger scale with further improvements.

6.2 Future Scope

The system can be improved by increasing the power capacity to support real electric vehicles. Wireless charging efficiency can be enhanced for better performance and longer range. Additional features like mobile control and advanced monitoring can be added.this project can also be expanded for use in real world smart charging station.

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